

State-Dependent tACS Effects Reveal the Potential Causal Role of Prestimulus Alpha Traveling Waves in Visual Contrast Detection

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The intricate relationship between prestimulus alpha oscillations and visual contrast detection variability has been the focus of numerous studies. However, the causal impact of prestimulus alpha traveling waves on visual contrast detection remains largely unexplored. In our research, we sought to discern the causal link between prestimulus alpha traveling waves and visual contrast detection across different levels of mental fatigue. Using electroencephalography alongside a visual detection task with 30 healthy adults (13 females; 17 males), we identified a robust negative correlation between prestimulus alpha forward traveling waves (FTWs) and visual contrast threshold (VCT). Inspired by this correlation, we utilized 45/–45° phase-shifted transcranial alternating current stimulation (tACS) in a sham-controlled, double-blind, within-subject experiment with 33 healthy adults (23 females; 10 males) to directly modulate these alpha traveling waves. After the application of 45° phase-shifted tACS, we observed a substantial decrease in FTW and an increase in backward traveling waves, along with a concurrent increase in VCT, compared with the sham condition. These changes were particularly pronounced under a low fatigue state. The findings of state-dependent tACS effects reveal the potential causal role of prestimulus alpha traveling waves in visual contrast detection. Moreover, our study highlights the potential of 45/–45° phase-shifted tACS in cognitive modulation and therapeutic applications.

Key words: alpha oscillations; mental fatigue; state-dependent effect; tACS; traveling waves; visual contrast detection

Significance Statement

Visual contrast detection, despite consistent stimuli, frequently exhibits variability. This variability has been linked to prestimulus alpha neural oscillations in prior studies. Recently, there has been increased interest in exploring large-scale alpha traveling waves and their connection with visual processing. Yet, the role of these traveling waves in visual contrast detection remains unclarified. Through a combination of visual detection tasks, electroencephalography data analysis, and 45/–45° phase-shifted transcranial alternating current stimulation, our study elucidates how prestimulus alpha traveling waves exert a potential causal influence on visual contrast detection.

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Introduction

Perceptions of consistent stimuli can exhibit remarkable variability, as individuals may detect a near-threshold visual contrast stimulus in one trial, while missing it in another. This perceptual variability has been associated with prestimulus (or spontaneous) neural oscillations, particularly within the alpha band (8–13 Hz; Samaha et al., 2020). A well-established correlation exists between reduced prestimulus alpha power (AP) and improved visual contrast detection (Samaha et al., 2020; Balestrieri and Busch, 2022; Wei et al., 2022). This relationship is thought to

stem from diminished AP, predominantly in the occipital region, potentially heightening the baseline excitability of the sensory systems (Samaha et al., 2020; Zhou et al., 2021; Iemi et al., 2022). Additionally, alpha oscillations exhibit a capacity for spatial propagation across cortical regions, forming “traveling waves” (Zhang et al., 2018; Alamia and VanRullen, 2019; Halgren et al., 2019). Recent findings suggest that these alpha traveling waves may support both top-down and bottom-up processing in visual tasks, associating backward traveling waves (BTWs) from frontal to occipital regions with top-down attentional mechanisms and forward traveling waves (FTWs) from occipital to frontal regions with bottom-up sensory processing (Alamia et al., 2023). Despite these advancements, the specific influence of prestimulus alpha traveling waves on visual contrast detection remains poorly understood.

While neuroimaging techniques like functional MRI and electroencephalogram (EEG) have shed light on the neural underpinnings of many cognitive processes, they often fall short in establishing causal links between neural activity and cognition (Ekhtiari et al., 2022; Siddiqi et al., 2022). Transcranial alternating current stimulation (tACS) presents a valuable approach to address this limitation, offering a means to directly modulate neural oscillations and investigate their influence on the behaviors (Beliaeva et al., 2021; Grover et al., 2021, 2023; Wischniewski et al., 2023a). Notably, recent studies in nonhuman primates have revealed that phase-shifted tACS with a 45 or -45° anterior-posterior phase difference can induce backward or forward propagating electric fields in the brain (Aleksichuk et al., 2019). Such findings pave the way for

investigating the causal impact of prestimulus traveling waves on visual contrast detection. Furthermore, it is important to consider that when using tACS to provide causal evidence, the effects of such stimulation are closely linked to specific brain states. This emerging consensus, highlighted in recent literature (Bradley et al., 2022), acknowledges that factors such as arousal (Scangos et al., 2021), sleepiness (Steinmann et al., 2022), and ongoing oscillations (Krause et al., 2022) can significantly influence the neural stimulation effects. However, the state-dependent effects of $45/-45^\circ$ phase-shifted tACS on prestimulus traveling waves and visual contrast detection are still largely unknown.

Our study aimed to elucidate the causal relationship between prestimulus alpha traveling waves and visual contrast detection through the state-dependent effects of $45/-45^\circ$ phase-shifted tACS. The first experiment combined EEG data collection with a visual detection task, revealing a negative correlation between prestimulus alpha FTW and visual contrast threshold (VCT). Building on this, the second experiment employed a sham-controlled, double-blind, within-subject design, integrating $45/-45^\circ$ phase-shifted tACS with pre- and poststimulation EEG during an identical visual detection task to modulate traveling waves and VCT. The results demonstrated that 45° phase-shifted tACS, in comparison with the sham condition, leads to a substantial decrease in FTW and an increase in BTW, along with a concurrent increase in VCT, particularly under a low fatigue state (see a schematic summary in Fig. 1). These results underscore a potential causal link between prestimulus alpha traveling waves and visual contrast detection.

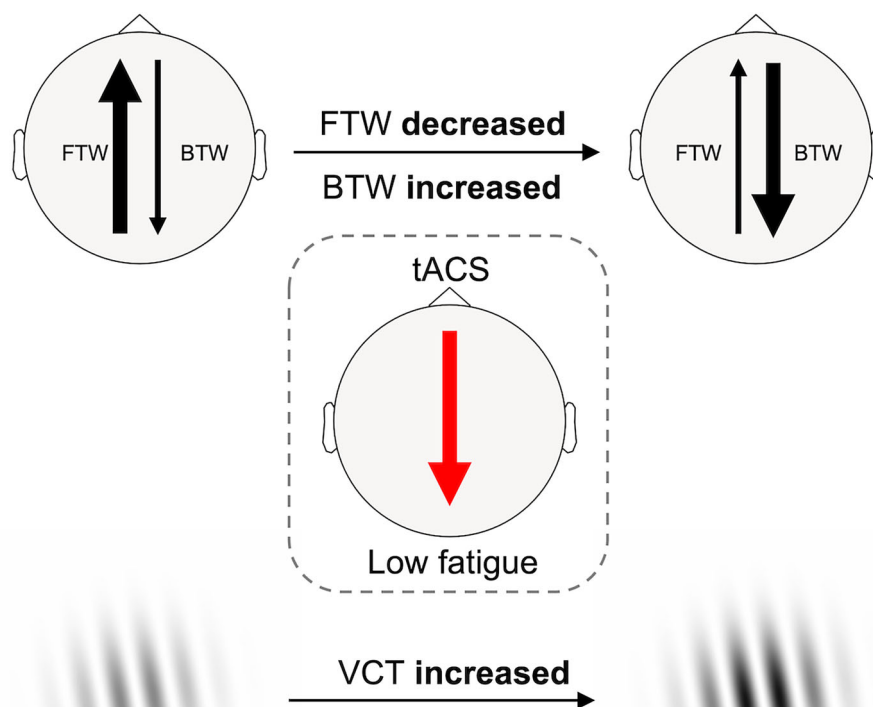


Figure 1. A schematic summary of our key findings. The 45° phase-shifted tACS generated backward propagating electric fields (red down arrow). It decreased FTW (black up arrow), increased BTW (black down arrow), and concurrently increased VCT under low fatigue, thereby establishing a potential causal link between prestimulus alpha traveling waves and visual contrast detection.

Materials and Methods

Experiment 1: EEG experiment

Participants. A total of 30 participants (13 females; age, 22.2 ± 1.5 years) were included in the study after excluding 8 participants due to task misunderstanding or excessive EEG artifacts. Participants were on-campus students with normal or corrected-to-normal vision and right-handedness. Before the study, participants were instructed to obtain sufficient rest and abstain from caffeine or alcohol consumption. All participants provided informed consent. The study was approved by the local ethics committee at Shenzhen University.

Experimental design. We reanalyzed data from a previous EEG study (Wei et al., 2022). The experiment was conducted in a dark and acoustically isolated chamber. The visual stimuli, near-threshold Gabor patches, were presented on a 21 in LCD monitor (100 Hz refresh rate, 60 cm distance) against a black background. Participants responded by pressing buttons after a 0.4 s delay, indicating perception when the fixation dot turned into a question mark. Feedback was provided by a green (correct) or red (incorrect) dot displayed for 0.2 s. A blank screen for blinking and an interstimulus interval [3,500, 4,500] ms followed (Fig. 2A). The experiment consisted of 16 blocks of 40 trials each (Fig. 2B). Additional details regarding the experimental setup and procedures were comprehensively described in Wei et al., (2022). Considering feedback from our pilot study where participants frequently experienced fatigue, we incorporated a measure of self-reported fatigue for future potential analysis. Before and after each block, participants rated their fatigue on a seven-point scale, with 1 being “no fatigue,” 4 as “moderate fatigue,” and 7 indicating “extreme fatigue.”

We used the QUEST algorithm (Watson and Pelli, 1983) to dynamically adjust the Gabor stimulus contrast in stimulus-present trials, aiming to maintain a 50% hit rate. The QUEST was continuously run throughout the experiment, and the prior was updated using results from the previous block. For the first block, the QUEST algorithm was initialized with the following priors: beta, 3.5; gamma, 0; delta, 0.01; pThreshold, 0.5; tGuess, 0.3; and tSD, 0.25. It is important to note that both tGuess and tSD, initially set in VCT units, were subject to log transformation in the QUEST implementation. The stimulus contrast values for each block were fitted using a cumulative Gaussian function, from which the VCT was estimated as the function's mean (Tomassini et al., 2020). Specifically, participants' responses to varying contrast levels of the Gabor patches were recorded, including both “hit” and “miss”. We fitted a cumulative Gaussian function to these responses to estimate the VCT for each participant. The function used was as follows:

$$\Phi(x) = \frac{1}{2} \left[1 + \operatorname{erf} \left(\frac{x - \mu}{\sigma\sqrt{2}} \right) \right] \quad (1)$$

where $\Phi(x)$ represents the cumulative Gaussian function, x is the stimulus contrast, μ is the mean of the distribution (corresponding to the estimated VCT), and σ is the standard deviation, representing the spread or variability of the participant's responses. The error function (erf) quantifies the probability that a random variable from a normal distribution with a mean of 0 and a standard deviation of 1 falls within a specified range.

EEG recording and preprocessing. EEG signals were acquired from a 64-channel EEG electrode system (EasyCap) and an EEG amplifier (BrainAmp, Brain Products) following the international 10–10 system. Electrode impedances were kept below 10 k Ω . The raw EEG data were preprocessed with the EEGLAB toolbox (Delorme and Makeig, 2004). All data were resampled to 250 Hz, followed by a bandpass FIR filtering from 0.1 to 80 Hz. The data were off-line rereferenced to the mean of both mastoid sites and segmented into epochs [−3,500, 1,500] ms relative to the stimulus onset. Visual artifact removal was followed by independent component analysis to identify and isolate nonbrain activities.

Traveling wave analysis. We implemented the traveling wave analysis method as detailed in Alamia et al. (2020, 2023) to assess prestimulus alpha traveling waves—specifically the FTW (occipital to frontal regions) and BTW (frontal to occipital regions; Fig. 3C). Within the [−3,500, 0] ms prestimulus interval, we conducted a 2D fast Fourier transform (2D-FFT) on time-electrode EEG data spanning midline electrodes (Fz, FCz, Cz, CPz, Pz, POz, Oz). Alpha band (8–13 Hz) peaks in the lower- and upper-right quadrants of the 2D-FFT output were identified as the primary indicators of the forward and backward wave power (FW and BW; Fig. 3A). For baseline comparisons, the electrode order was shuffled 100 times to average the peak values of the shuffle-derived forward and backward waves (SFW and SBW; Fig. 3B). The final power of the traveling waves in decibels was quantified using the following formula:

$$\text{FTW} = 10 \times \log_{10} \frac{\text{FW}}{\text{SFW}_{\text{mean}}}; \quad \text{BTW} = 10 \times \log_{10} \frac{\text{BW}}{\text{SBW}_{\text{mean}}} \quad (2)$$

Importantly, the estimation of alpha FTW/BTW using this technique accounts for the variability of alpha peak frequencies (APFs) across the electrodes of interest. To illustrate, we simulated FTWs propagating across seven electrodes under three scenarios: (1) APFs exhibit minor variations across electrodes (Fig. 4A); (2) APFs are present in several but not all electrodes (Fig. 4B); and (3) APFs are not detectable (Fig. 4C). In all scenarios, traveling waves oscillated within the 5–15 Hz frequency band, with a phase offset of $\pi/5$ applied to the alpha band frequencies between adjacent electrodes (see details in the code repository). Regardless of the presence or absence of APFs, FW/BW and SFW/SBW in the alpha band were effectively identified. Thus, FTW and BTW can be computed using Equation 2, demonstrating the method's robustness to APFs across electrodes.

Statistical analysis of correlations between prestimulus alpha traveling waves and VCT. Data samples of FTW and BTW were considered outliers and removed if their deviation from the median exceeded 2.5 times the median absolute deviation within each block for each participant (Leys et al., 2013). The remained FTW and BTW samples were averaged within each block and then standardized within each participant to reduce multicollinearity. Bayesian linear mixed models (Franke and Roettger, 2019) were fitted to the VCT as a function of FTW or BTW (treated as fixed effects), using the Stan modeling language (Carpenter et al., 2017) and the package *brms* (Bürkner, 2017). The Bayesian framework was utilized due to its advantages in

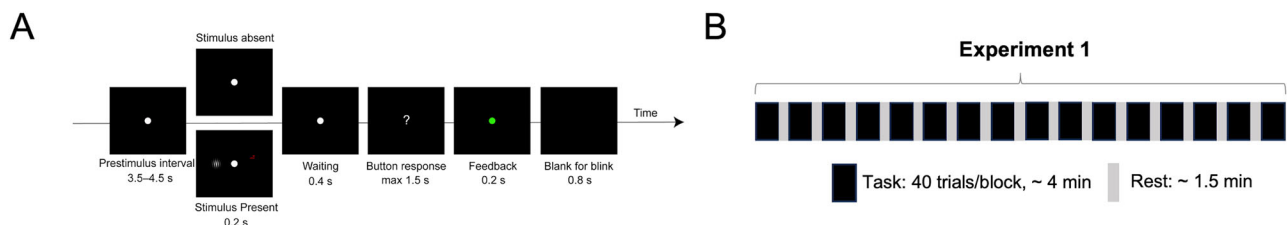


Figure 2. Experiment 1 setup. **A**, A near-threshold visual stimulus appeared on either the left or right side (50% probability) of a central fixation dot for 0.2 s in 60% of trials. Participants responded by button presses to indicate stimulus perception when the fixation dot changed to a question mark after a 0.4 s interval. Feedback was provided by a green (correct) or red (incorrect) dot for 0.2 s, followed by a blank screen for blinking. The next trial started after 3.5–4.5 s. **B**, Experiment 1 comprised 16 blocks, each consisting of 40 trials.

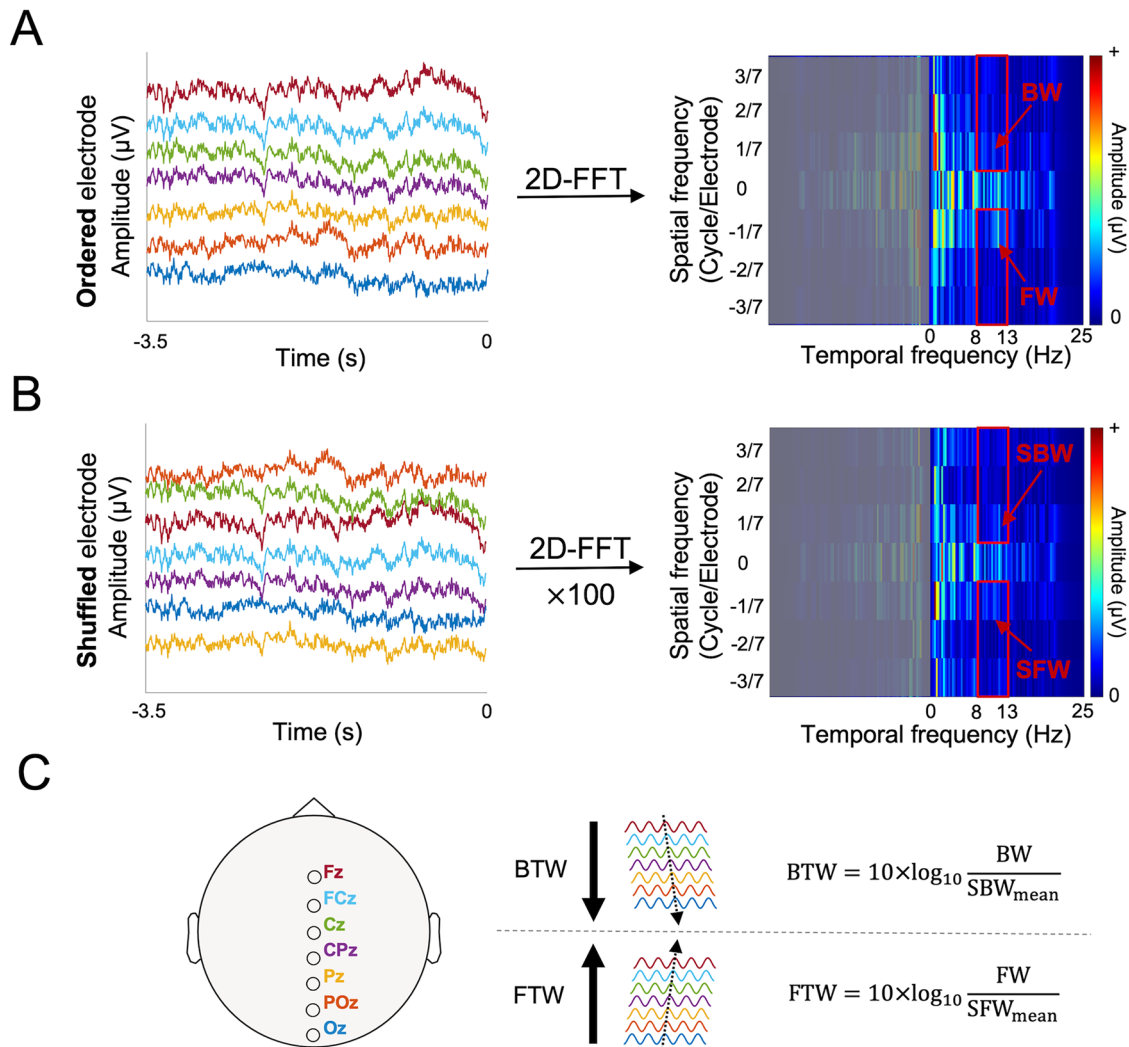


Figure 3. Traveling waves analysis. **A**, EEG data within the $[-3,500, 0]$ ms prestimulus interval across midline electrodes (Fz, FCz, Cz, CPz, Pz, POz, Oz) were subjected to 2D-FFT. Maximal values within the alpha band's lower- and upper-right quadrants, indicated by red rectangles, reflect the power of FW and BW, respectively. **B**, To establish baseline measures, the order of electrodes was shuffled 100 times, each iteration capturing the maximal forward and backward wave values (SFW and SBW). **C**, The power of the traveling waves (BTW/FTW) were calculated in decibels by normalizing BW/FW with the mean of SBW/SFW. The black down and up arrows depict the respective propagation directions of BTW and FTW.

addressing issues of nonconvergence and allowing the fitting of maximal varying effect structures (Nalborczyk et al., 2019). In line with our previous study (Wei et al., 2022), the model syntax was $VCT \sim 1 + FTW / BTW + (1|Participant) + (1|Block)$. Here, participant and block were treated as random effects, and the default priors of the *brms* package were applied. Four sampling chains were run for 3,000 iterations, with the first 1,000 iterations serving as the warm-up phase. We reported the estimated slopes ($E(\mu_{FTW})$, $E(\mu_{BTW})$) under the posterior distribution and their corresponding 95% highest probability density (HPD). Evidence supporting the correlation is considered compelling if the posterior probability (P) exceeds 97.5% and the 95% HPD excludes 0 (Franke and Roettger, 2019; Cowan et al., 2022). Effect sizes δ_i are calculated by the estimated values divided by the pooled standard deviation (the square root of the sum of all variance components; Hedges, 2007; Nalborczyk et al., 2019).

Experiment 2: a 45/–45° phase-shifted tACS experiment

Participants. Thirty-three participants (23 females; age, 20.3 ± 2.1 years) were included in the study after excluding five individuals due to task misunderstanding, excessive EEG artifacts, or discomfort during the stimulation. The inclusion criteria mirrored those of Experiment 1, with additional exclusions for left-handedness, history of seizures or head injuries, use of psychotropic medication, presence

of metal implants inside the head, implanted electronic devices, current psychiatric disorders, tinnitus, and participation in other neuromodulation experiments within the last 3 months.

Experimental design. This experiment aimed to modulate prestimulus alpha traveling waves (FTW and BTW) and assess their effects on VCT. We replicated the visual detection task from Experiment 1 (Fig. 2A), including data collection on FTW, BTW, and VCT. After an initial familiarization phase, the experiment was segmented into seven blocks (Fig. 5A), with Block 1 serving as a baseline. Blocks 2–4 and 5–7 were designated for during-stimulation and poststimulation sessions, respectively. EEG recordings were taken in the prestimulation and poststimulation sessions. Consistent with Experiment 1, fatigue ratings were collected using a seven-point questionnaire before and after each block, with the mean score representing the prevalent fatigue state for each block. Data samples of FTW, BTW, and VCT were classified based on these fatigue ratings, employing a bifurcation threshold of 4 on the seven-point scale to distinguish between low (<4) and high (≥ 4) fatigue levels (Constans et al., 1999; de Rezende and de Medeiros, 2022). While feedback from Experiment 1 highlighted consistent fatigue experiences, a primary rationale for this measure in Experiment 2 stemmed from the recognized interplay between brain states and stimulation effects (Bradley et al., 2022). Specifically, conditions like arousal

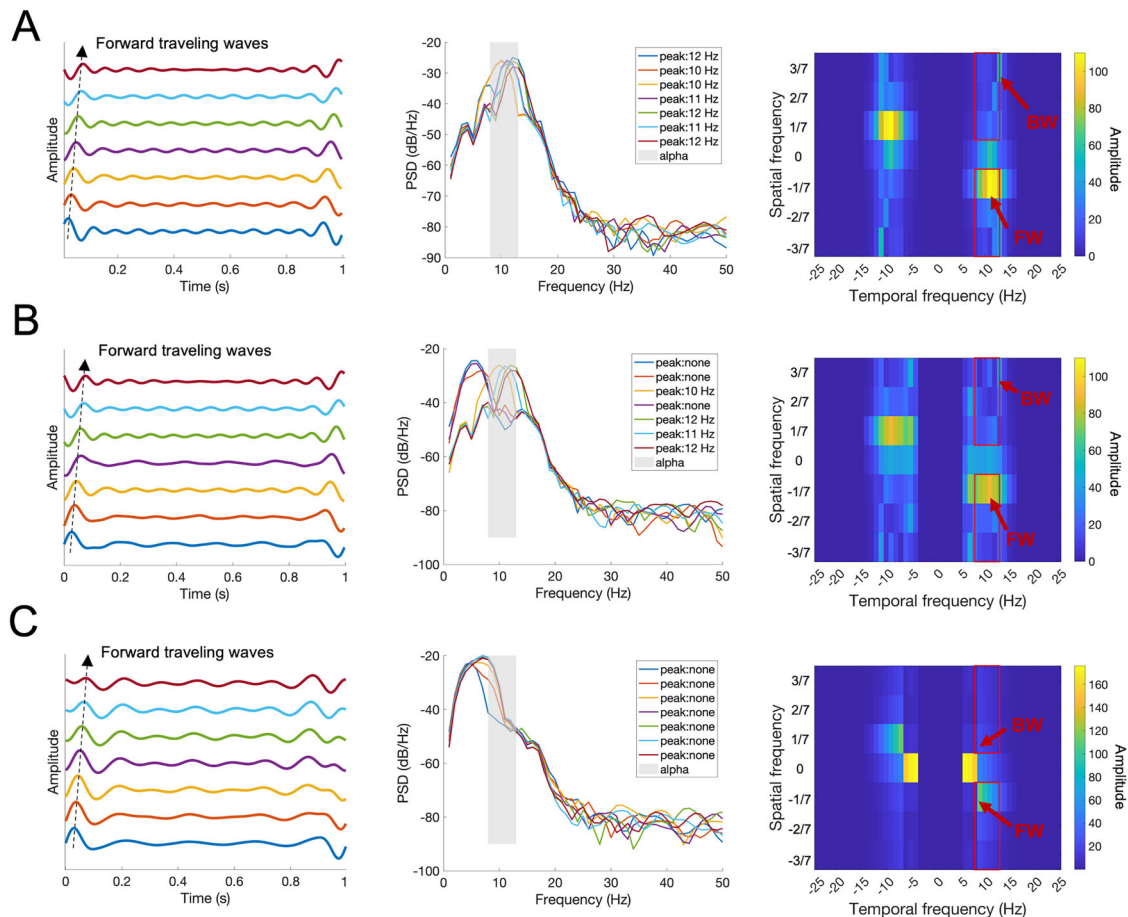


Figure 4. Robustness of traveling waves analysis. **A**, Left, Simulated FTWs propagated across seven electrodes with slight variations in APFs. Middle, Power spectral density (PSD) profiles were generated via Welch’s method, showing the slight variability of APFs among channels (10–12 Hz). Right, The 2D-FFT of the ordered electrode signals effectively identifies FW and BW within the alpha band, demarcated by red rectangles. **B**, Left, Simulated FTWs propagated across seven electrodes, with some displaying distinct APFs, while others did not. Middle, PSD analysis shows electrodes with diverse alpha peaks, some with identifiable peaks at specific frequencies and others without a clear peak, denoted as “none.” Right, FW and BW were effectively identified. **C**, Left, Simulated FTWs propagated across seven electrodes in the absence of APFs. Middle, PSD profiles indicate no discernible alpha peaks across all electrodes. Right, FW and BW can be identified even when APFs are absent.

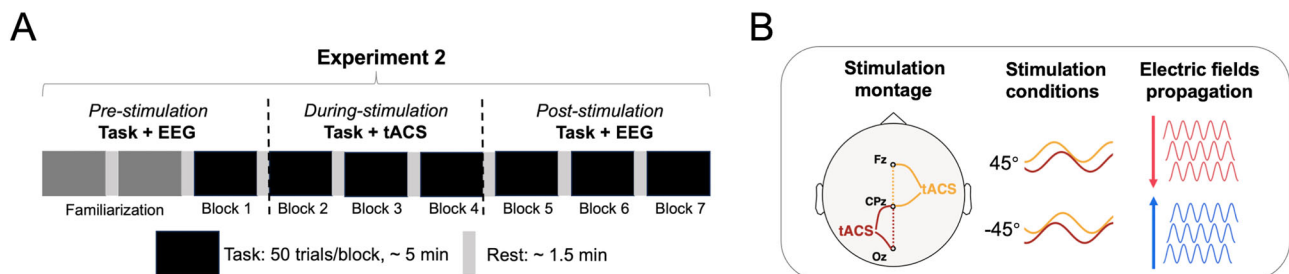


Figure 5. Experiment 2 setup and 45/–45° phase-shifted tACS. **A**, The visual detection task used in Experiment 1 was adapted for Experiment 2, involving seven blocks and three sessions (pre-, during-, and post-stimulation). **B**, Three stimulation conditions (45° phase-shifted tACS, –45° phase-shifted tACS, sham) were administered on separate days with an interval of 3–7 d, following a sham-controlled, double-blinded, within-subject design. Phase-shifted tACS, with phase differences of either 45° or –45° between the Fz-CPz and Oz-CPz loops, produced either backward or forward propagating electric fields. The red down arrow indicates the backward direction (from frontal to occipital regions), while the blue up arrow represents the forward direction.

and sleepiness, which are closely linked with fatigue (Bafna and Hansen, 2021), have been documented to significantly influence electrical stimulation responses (Scangos et al., 2021; Steinmann et al., 2022). Consequently, we postulated state-dependent modulatory effects during the present experiment, namely, stimulations may have different modulatory effects under different fatigue levels.

A 45/–45° phase-shifted tACS. Phase-shifted tACS was administered using a high-definition transcranial electrical stimulator

(Soterix Medical) with an $M \times N$ 33-channel configuration. Three 12-mm-diameter Ag/AgCl ring electrodes filled with conductive gel were placed over Oz and Fz locations, with a common reference electrode at CPz (Fig. 5B). Sinusoidal stimulation waveforms with a phase difference of either 45° or –45° between the two electrical loops were used to generate electric fields that propagated backward (from frontal to occipital regions) or forward (from occipital to frontal regions) within the brain (Aleksichuk et al., 2019). Our hypothesis was that 45° phase-shifted tACS would decrease FTW and increase BTW,

thereby raising VCT, and the opposite for -45° phase-shifted tACS. The stimulation frequency (10.6 ± 0.6 Hz) was individually determined in the familiarization period as the center of gravity frequency within the 8–13 Hz band (Hooper, 2005; Bazanova and Vernon, 2014), defined by the following:

$$\text{IAF} = \frac{\sum f_i \times P_i}{\sum P_i} \quad (3)$$

where f_i are the frequencies 8, 9, 10, 11, 12, and 13 Hz and P_i is the power at f_i calculated by Welch's method. The stimulation intensity was calibrated individually below phosphene and discomfort levels (0.37 ± 0.07 mA peak to peak) using a staircase procedure similar to Vossen et al. (2015).

The study adhered to a randomized, double-blind, sham-controlled design, encompassing three stimulation conditions: 45° phase-shifted tACS, -45° phase-shifted tACS, and sham. One experimenter administered the stimulation, while the participant and another experimenter responsible for data collection were blinded to the stimulation condition. Each participant underwent all three conditions with a 3–7 d interval. All stimulation conditions included a 10 s fade-in/fade-out period, while active stimulation conditions ($45/-45^\circ$) were applied during entire Blocks 2–4 (not including rest), and the sham stimulation was limited in the first 30 s of Blocks 2–4. Participants completed a questionnaire to rate common adverse effects after each stimulation (Matsumoto and Ugawa, 2017). A repeated-measure ANOVA showed no significant differences in adverse effect ratings across three stimulation conditions ($F_{(2, 64)} = 1.048$; $p = 0.356$).

EEG recording and processing. EEG data collection (across Blocks 1, 5–7) and preprocessing procedures closely followed Experiment 1. Prestimulus FTW and BTW were computed using the same method as in Experiment 1.

Statistical analysis of state-dependent tACS effects on FTW, BTW, and VCT. Bayesian linear mixed models, identical in parameters (e.g., priors, chains) to those in Experiment 1, were employed for statistical analysis. Initially, we examined if different stimulation conditions influenced fatigue ratings, potentially confounding the categorization of low/high fatigue (Jagannathan et al., 2022). This was followed by an analysis of how during- and post-stimulation VCT varied across blocks in the different stimulation conditions under different fatigue levels (low/high). Lastly, we focused on investigating the aftereffects (i.e., Blocks 5–7), as EEG data during stimulation was unavailable due to substantial artifacts. We analyzed how poststimulation FTW, BTW, and VCT differed between the three stimulation conditions under different fatigue levels. Baseline measurements from Block 1 were incorporated into the models as the covariates, and the sham condition at Block 1 always served as the reference level. The results were evaluated by calculating the posterior probability of the contrast difference, δ , between estimated parameters corresponding to active and sham conditions. The models formulated hypotheses stating that $\delta \neq 0$. The evidence supporting these hypotheses was considered compelling if the posterior probability (P) exceeds 97.5% and 95% HPD does not include 0 (Franke and Roettger, 2019; Cowan et al., 2022). The specific models employed were defined as follows:

1. During- and after-effects of stimulation condition on fatigue ratings (Fig. 7A): Fatigue ratings were modeled as a function of baseline measurements, stimulation condition, block, and their interactions, with individual participants as random effects. The model syntax was as follows:

$$\text{Fatigue ratings} \sim 1 + \text{Baseline} + \text{Condition} \times \text{Block} + (1 + \text{Condition} \times \text{Block} | \text{Participant}).$$

2. During- and after-effects of stimulation condition on VCT depending on the fatigue level (Fig. 7B,C): VCT was modeled to analyze the impact of the fatigue level, the stimulation condition, block, and their interactions on visual contrast detection. For the random effects

part, we did not include the interaction of the block due to the increased complexity and implementation failure when running. The model was formulated as follows:

$$\text{VCT} \sim 1 + \text{Baseline} + \text{Fatigue level} \times \text{Condition} \times \text{Block} + (1 + \text{Fatigue level} \times \text{Condition} | \text{Participant}).$$

3. After-effects of stimulation condition on FTW, BTW, and VCT depending on the fatigue level (Fig. 8):

The model for FTW, BTW, and VCT focused on the after-effects of the fatigue level, stimulation condition, and their interactions. Unlike the previous models, block information was not considered, as the emphasis was on evaluating the overall after-effects across the post-stimulation period. The model syntax for each dependent variable was as follows:

$$\text{FTW/BTW/VCT} \sim 1 + \text{Baseline} + \text{Fatigue level} \times \text{Condition} + (1 + \text{Fatigue level} \times \text{Condition} | \text{Participant}).$$

Reevaluation of correlation in Experiment 1

Given the varying stimulation effects across different fatigue states observed in Experiment 2, it became imperative to reevaluate whether the correlations between FTW/BTW and VCT in Experiment 1 were similarly predominated by a certain fatigue level. To accomplish this, we categorized data on FTW, BTW, and VCT into low and high fatigue levels, mirroring the approach in Experiment 2. A Bayesian linear mixed model, maintaining consistency in parameters with previous analyses, was employed. The model assessing the correlations between FTW/BTW and VCT was formulated as follows:

$$\text{VCT} \sim 1 + \text{Fatigue level} \times \text{FTW/BTW} + (1 + \text{Fatigue level} | \text{Participant}) + (1 + \text{Fatigue level} | \text{Block}).$$

It is important to note that the interaction between the fatigue level and FTW/BTW was not included in the model's random effects. This decision stems from the design of the preceding model, $\text{VCT} \sim 1 + \text{FTW/BTW} + (1 | \text{Participant}) + (1 | \text{Block})$, which did not incorporate FTW/BTW in its random effects framework.

Control analysis of prestimulus AP

In light of the robust association between visual contrast detection and prestimulus AP in prior studies (Samaha et al., 2020), we conducted a control analysis to exclude the possibility that the observed variations in VCT were attributable to alterations in AP, potentially resulting from the tACS effects.

Custom MATLAB scripts were used to calculate AP averaged across occipital electrodes Oz, PO3, PO4, O1, and O2. The EEG data were segmented into intervals spanning $[-3,500, 500]$ ms and then converted into time–frequency representations via complex Morlet wavelets following the same parameters as Wei et al. (2022). AP was obtained by squaring the magnitude of the complex result and averaging them within 8–13 Hz, $[-3,500, -150]$ ms. The -150 ms accounted for the windowing effect (Iemi et al., 2017).

The methodology was consistent with previous analyses. First, the correlation between AP and VCT was analyzed using the following model: $\text{VCT} \sim 1 + \text{Fatigue level} \times \text{AP} + (1 + \text{Fatigue level} | \text{Participant}) + (1 + \text{Fatigue level} | \text{Block})$. Second, the after-effects of stimulation condition on AP were analyzed using the following model: $\text{AP} \sim 1 + \text{Baseline} + \text{Fatigue level} \times \text{Condition} + (1 + \text{Fatigue level} \times \text{Condition} | \text{Participant})$.

Additionally, given the reported association between AP and alpha traveling waves (Alamia et al., 2023), we further analyzed the correlations between AP and FTW/BTW using the following models: $\text{FTW/BTW} \sim 1 + \text{Fatigue level} \times \text{AP} + (1 + \text{Fatigue level} | \text{Participant}) + (1 + \text{Fatigue level} | \text{Block})$.

Results

Experiment 1: FTW negatively correlated with VCT

A substantial negative correlation emerged between FTW and VCT, indicated in Figure 6B ($E(\mu_{\text{FTW}}) = -0.392$, HPD = $[-0.766, -0.020]$, $P(\mu_{\text{FTW}} < 0) = 0.978$, $\delta_i = -0.126$,

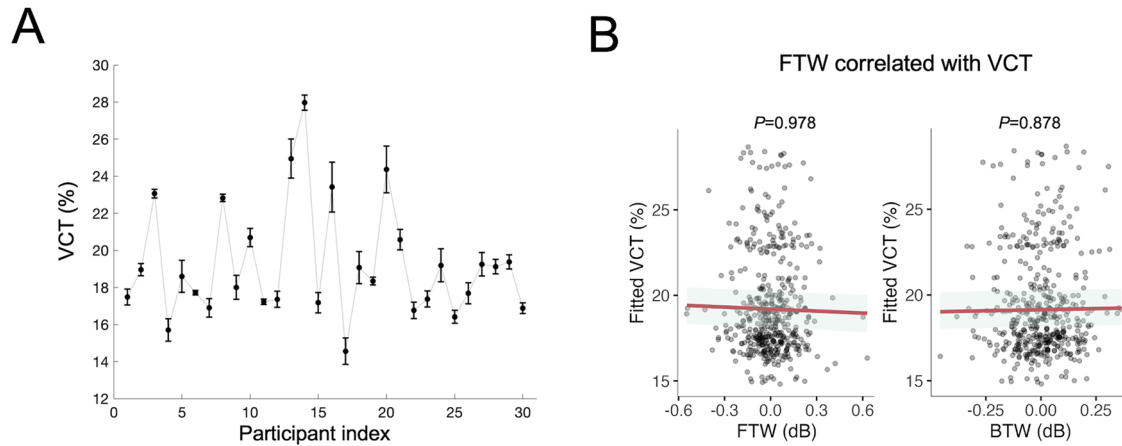


Figure 6. Behavioral and correlation results in Experiment 1. **A**, Variability in VCT across blocks for individual participants, depicted with error bars indicating standard deviation. **B**, Correlation analysis between FTW/BTW and VCT. FTW negatively correlated with VCT (left panel). BTW did not correlate with VCT statistically (right panel). Observed data are represented by dots, the regression line by a thick red line, and the 95% HPD by the shaded green area. The evidence is considered compelling if a posterior probability (P) exceeds 97.5%, accompanied by a 95% HPD that does not include 0.

HPD = $[-0.248, -0.004]$). However, no robust correlation was found between BTW and VCT ($E(\mu_{\text{BTW}}) = 0.259$, HPD = $[-0.167, 0.702]$, $P(\mu_{\text{BTW}} > 0) = 0.878$, $\delta_t = 0.082$, HPD = $[-0.067, 0.217]$). It is crucial to mention that both Figure 6B and the regression results for rest illustrate the overall trend in the data considering the variability between participants and between blocks. In addition, VCT demonstrated a relatively small standard deviation for each participant (Fig. 6A). Therefore, the subtle slopes seen in Figure 6B, while appearing to be minimal, are indeed significant: the modest changes in the regression lines can reflect considerable shifts in subjective contrast perception.

Experiment 2: 45° phase-shifted tACS decreased FTW, increased BTW, and concurrently increased VCT depending on the fatigue level

Following the observed negative correlation between FTW and VCT in Experiment 1, we employed 45/−45° phase-shifted tACS to establish causality between traveling waves (FTW/BTW) and VCT in Experiment 2. Our initial analysis revealed no significant influence of stimulation conditions on fatigue ratings, ensuring the independence of fatigue from the stimulation conditions (Fig. 7A).

We then explored the state-dependent tACS effects on VCT in the during- and post-stimulation sessions (Blocks 2–7). Under low fatigue, 45° phase-shifted tACS increased VCT from Blocks 5 to 7 compared with sham (Fig. 7B; $E(\mu_{\text{Sham}-45^\circ, \text{Block 5}}) = -0.624$, HPD = $[-1.085, -0.154]$, $P(\mu_{\text{Sham}-45^\circ, \text{Block 5}} < 0) = 0.994$, $\delta_t = 0.353$, HPD = $[0.147, 0.588]$; $E(\mu_{\text{Sham}-45^\circ, \text{Block 6}}) = -0.776$, HPD = $[-1.226, -0.304]$, $P(\mu_{\text{Sham}-45^\circ, \text{Block 6}} < 0) = 0.999$, $\delta_t = 0.434$, HPD = $[0.217, 0.682]$; $E(\mu_{\text{Sham}-45^\circ, \text{Block 7}}) = -0.920$, HPD = $[-1.398, -0.460]$, $P(\mu_{\text{Sham}-45^\circ, \text{Block 7}} < 0) = 0.999$, $\delta_t = 0.506$, HPD = $[0.262, 0.755]$). However, tACS did not affect VCT across blocks under high fatigue ($P_s < 0.975$; Fig. 7C).

Further, we investigated the state-dependent tACS effects on FTW, BTW, and VCT in the poststimulation session (Blocks 5–7). Under low fatigue, 45° phase-shifted tACS decreased FTW (Fig. 8A; $E(\mu_{\text{Sham}-45^\circ, \text{FTW}}) = 0.075$; HPD = $[0.011, 0.142]$; $P(\mu_{\text{Sham}-45^\circ, \text{FTW}} > 0) = 0.988$; $\delta_t = -0.275$; HPD = $[-0.533, -0.041]$). Both 45° and −45° phase-shifted tACS increased BTW compared with sham (Fig. 8B; $E(\mu_{\text{Sham}-(-45^\circ), \text{BTW}}) = -0.049$,

HPD = $[-0.087, -0.013]$, $P(\mu_{\text{Sham}-(-45^\circ), \text{BTW}} < 0) = 0.992$, $\delta_t = 0.212$, HPD = $[0.047, 0.400]$; $E(\mu_{\text{Sham}-45^\circ, \text{BTW}}) = -0.050$, HPD = $[-0.097, -0.003]$, $P(\mu_{\text{Sham}-45^\circ, \text{BTW}} < 0) = 0.982$, $\delta_t = 0.217$, HPD = $[0.008, 0.437]$). The 45° phase-shifted tACS increased VCT (Fig. 8C; $E(\mu_{\text{Sham}-45^\circ, \text{VCT}}) = -0.804$, HPD = $[-1.294, -0.353]$, $P(\mu_{\text{Sham}-45^\circ, \text{VCT}} < 0) = 1$, $\delta_t = 0.320$, HPD = $[0.126, 0.539]$). Under high fatigue, either FTW/BTW or VCT remained unaffected by 45/−45° phase-shifted tACS ($P_s < 0.975$; Fig. 8A–C).

Given the direct physiological impact of tACS (Beliaeva et al., 2021; Grover et al., 2021; Wischniewski et al., 2023a), it is plausible that 45° phase-shifted tACS directly affected FTW and BTW, leading to the observed increase in VCT, particularly under low fatigue.

Reevaluation of correlations: FTW–VCT correlation was predominated by the low fatigue state

Investigating the state dependence of the correlation found in Experiment 1, we discovered that the correlation between FTW and VCT was also predominated by the low fatigue state. As illustrated in Figure 9, under low fatigue, a robust negative correlation was observed between FTW and VCT ($E(\mu_{\text{FTW}; \text{Low}}) = -0.651$, HPD = $[-1.048, -0.235]$, $P(\mu_{\text{FTW}; \text{Low}} < 0) = 0.999$, $\delta_t = -0.214$, HPD = $[-0.360, -0.072]$), while no such correlation existed under high fatigue ($P(\mu_{\text{FTW}; \text{High}} > 0) = 0.518$). There was no pronounced difference of the estimated slopes between two fatigue levels ($P(\mu_{\text{FTW}; \text{Low}} - \mu_{\text{FTW}; \text{High}} < 0) = 0.946$). Regarding BTW, the slope was not robustly different from zero under either fatigue level ($P(\mu_{\text{BTW}; \text{Low}} > 0) = 0.942$; $P(\mu_{\text{BTW}; \text{High}} > 0) = 0.573$), and similarly, there was no compelling evidence showing the difference between the slopes under the two fatigue levels ($P(\mu_{\text{BTW}; \text{Low}} - \mu_{\text{BTW}; \text{High}} > 0) = 0.764$).

Control analysis: VCT variations cannot be attributed to the changes of prestimulus AP

To ensure that the observed variations in VCT were not attributable to alterations in AP, we conducted an analysis to explore both the correlation between VCT and AP in Experiment 1 and the potential effects of tACS on AP in Experiment 2. Experiment 1 revealed no correlations between AP and VCT (Fig. 10A; $P(\mu_{\text{AP}; \text{Low}} > 0) = 0.704$, $P(\mu_{\text{AP}; \text{High}} > 0) = 0.798$,

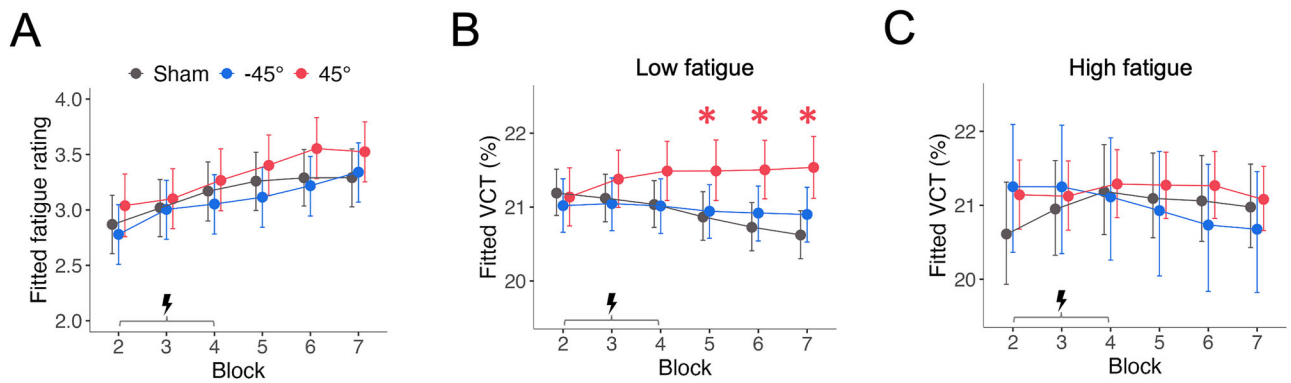


Figure 7. tACS effects on fatigue ratings and VCT across blocks in the during- and post-stimulation sessions of Experiment 2. **A**, Fatigue ratings remained unaffected by stimulation conditions, affirming fatigue as an independent variable for categorization. The lightning symbol denotes the during-stimulation session. **B**, Effects of phase-shifted tACS on VCT under low fatigue: 45° phase-shifted tACS increased VCT from Blocks 5 to 7 compared with the sham condition. The asterisk (*) indicates a posterior probability exceeding 97.5%, accompanied by a 95% HPD that does not include 0, providing compelling evidence. **C**, No effects of phase-shifted tACS on VCT under high fatigue.

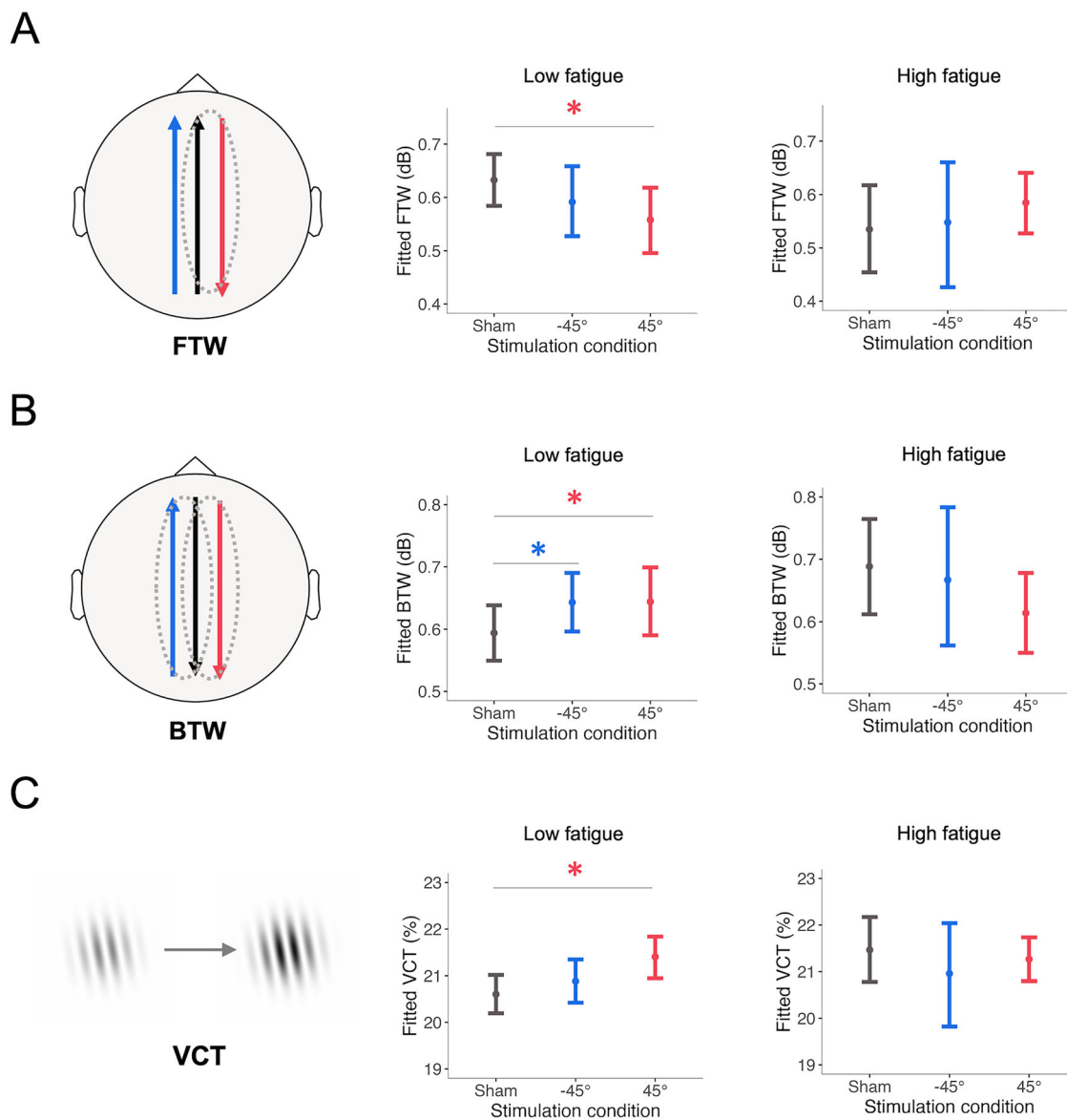


Figure 8. State-dependent tACS effects on FTW, BTW, and VCT in the poststimulation session of Experiment 2. **A**, The 45° phase-shifted tACS decreased FTW compared with the sham condition under low fatigue. In the left schematic summary, the red and blue arrows represent the backward and forward propagating electric fields generated by phase-shifted tACS, while the black up arrow denotes FTW. The dashed gray ellipse illustrates that the propagating electric fields influenced FTW. The asterisk (*) indicates a posterior probability exceeding 97.5%, accompanied by a 95% HPD that does not include 0, providing compelling evidence. **B**, The -45 and 45° phase-shifted tACS both increased BTW compared with the sham condition under low fatigue. The black down arrow denotes BTW. **C**, The 45° phase-shifted tACS increased VCT compared with the sham condition under low fatigue.

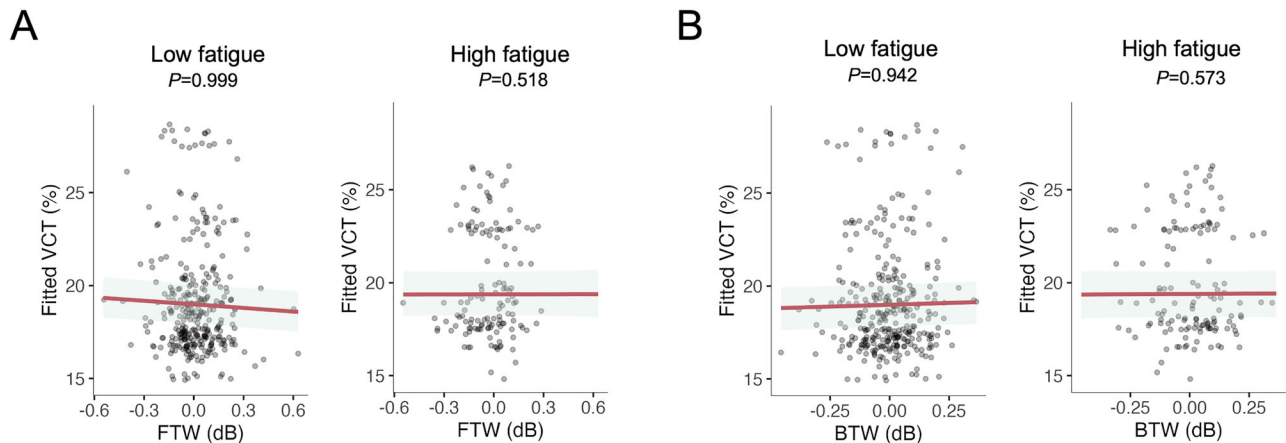


Figure 9. Reevaluation of correlations in Experiment 1. **A**, FTW correlated with VCT merely under low fatigue. Observed data are represented by dots, the regression line by a thick red line, and the 95% HPD by the shaded area. The evidence is considered compelling if a posterior probability (P) exceeds 97.5%, accompanied by a 95% HPD that does not include 0. **B**, BTW did not correlate with VCT under either fatigue level.

$P(\mu_{\text{AP}; \text{Low}} - \mu_{\text{AP}; \text{High}} < 0) = 0.636$). Experiment 2 demonstrated that -45° phase-shifted tACS decreased AP under low fatigue (Fig. 10B; $E(\mu_{\text{Sham}(-45^\circ); \text{Low}}) = 0.705$, HPD = [0.259, 1.122], $P(\mu_{\text{Sham}(-45^\circ); \text{Low}} > 0) = 0.999$, $\delta_t = -0.353$, HPD = [-0.584, -0.124]); however, 45° phase-shifted tACS did not affect AP (Fig. 10B; $E(\mu_{\text{Sham}45^\circ; \text{Low}}) = 0.470$, HPD = [-0.027, 1.021], $P(\mu_{\text{Sham}45^\circ; \text{Low}} > 0) = 0.963$, $\delta_t = -0.237$, HPD = [-0.501, 0.027]). Therefore, the changes in VCT by 45° phase-shifted tACS cannot be attributed to alterations in AP.

Furthermore, consistent with previous studies (Alamia et al., 2023), we found the negative correlations between AP and FTW under low and high fatigue (Fig. 10C; $E(\mu_{\text{FTW}; \text{Low}}) = -0.047$, HPD = [-0.065, -0.029], $P(\mu_{\text{FTW}; \text{Low}} < 0) = 1$, $\delta_t = -0.180$, HPD = [-0.271, -0.090]; $E(\mu_{\text{FTW}; \text{High}}) = -0.030$, HPD = [-0.057, -0.003], $P(\mu_{\text{FTW}; \text{High}} < 0) = 0.984$, $\delta_t = -0.163$, HPD = [-0.267, -0.093]; $P(\mu_{\text{FTW}; \text{Low}} - \mu_{\text{FTW}; \text{High}} < 0) = 0.852$) and positive correlations between AP and BTW only under low fatigue (Fig. 10D; $E(\mu_{\text{BTW}; \text{Low}}) = 0.036$, HPD = [0.020, 0.053], $P(\mu_{\text{BTW}; \text{Low}} > 0) = 1$, $\delta_t = 0.168$, HPD = [0.078, 0.252]).

Discussion

Our study sought to discern the causal link between prestimulus alpha traveling waves and visual contrast detection. The initial findings demonstrated a negative correlation between FTW and VCT. Moreover, the application of 45° phase-shifted tACS resulted in decreased FTW, increased BTW, and concurrent increased VCT under low fatigue. These state-dependent tACS effects suggest a potential causal link between prestimulus alpha traveling waves and visual contrast detection.

The potential causal link due to state-dependent tACS effects

To provide causal evidence, we began by analyzing correlations, following methodologies similar to those detailed in Polania et al. (2012). The detection of a correlative relationship led us to modulate traveling waves (FTW/BTW) using $45/-45^\circ$ phase-shifted tACS, based on the hypothesis that it would influence VCT. This hypothesis was grounded in the evidence that tACS directly interacts with neural physiology (Beliaeva et al., 2021; Grover et al., 2021; Wischniewski et al., 2023a) and could induce behavioral changes via these physiological changes. Our results confirmed this hypothesis, showing modulation in FTW, BTW, and VCT under a low fatigue state. It is important to recognize that while correlation does not imply causality, prompting our use of

neuromodulation, causality also does not necessarily suggest linear correlations (Cox and Wermuth, 2004), as evidenced in our observations of BTW. While we demonstrated a causal influence of FTW on VCT during low fatigue, the absence of tACS effects under high fatigue left the high fatigue state undetermined. Consequently, our findings contribute to the body of causal evidence, proposing a potential causal link that could be further elucidated with additional research.

Additionally, our control analysis affirmed the specificity of FTW/BTW effects by showing that the AP changes did not correspond to VCT variations. However, it was also shown that AP did not correlate with VCT, seemingly in contrast with the previous study (Balestrieri and Busch, 2022). This inconsistency was potentially due to differences in the prestimulus intervals examined. In our study, continuous stimulation affected the entire prestimulus interval, leading us to employ a longer interval than that used in the previous research for analyzing traveling waves and AP.

The potential causes of state-dependent tACS effects

Our results confirmed the anticipated state-dependent effects of tACS, revealing that FTW/BTW and VCT were predominantly modulated under low fatigue, the brain state closely associated with high arousal or low sleepiness (Bafna and Hansen, 2021). Analogous outcomes have been documented in prior research (Bradley et al., 2022). Scangos et al. (2021) reported a positive modulation effect for treating depression when intracranial electric stimulation was administered in the orbitofrontal cortex during states of high or neutral arousal. Similarly, Steinmann et al. (2022) posited that the aftereffects of tACS on alpha oscillations might go unnoticed owing to increased sleepiness.

However, the reasons certain brain states, such as low fatigue or high arousal, exhibit increased sensitivity to external stimulation remain somewhat elusive. In this context, relevant neuroimaging studies might offer some insights. A notable study highlighted an inverted U-shaped relationship between relative AP and arousal, revealing a declining trend in alpha oscillations—potentially indicating reduced sensitivity—during states of low arousal (Podvalny et al., 2021). Furthermore, heightened arousal states have been linked with an augmented excitation-inhibition balance across neuronal networks (Pfeffer et al., 2022). Regarding the role of fatigue in our discovered causal relationship, it is possible that the

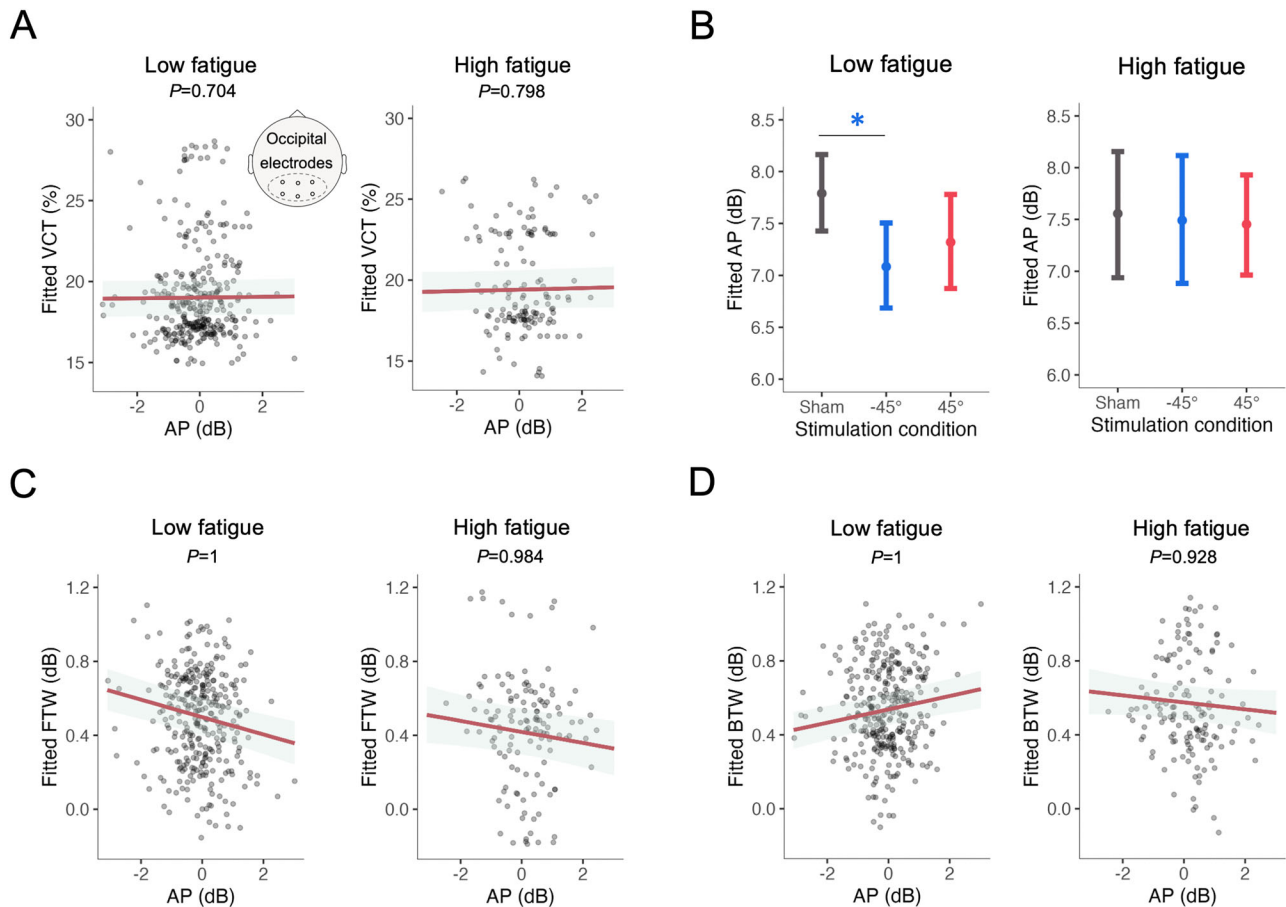


Figure 10. Control analysis of AP within the prestimulus interval [−3,500, −150] ms. **A**, AP did not correlate with VCT under either fatigue level in Experiment 1. AP values were averaged across six occipital EEG electrodes (P03, P04, P05, P06, P07, P08, shown by the inset topography). Observed data are represented by dots, the regression line by a thick red line, and the 95% HPD by the shaded area. The evidence is considered compelling if a posterior probability (P) exceeds 97.5%, accompanied by a 95% HPD that does not include 0. **B**, The −45° phase-shifted tACS decreased AP compared with the sham condition merely under low fatigue in Experiment 2. The asterisk (*) indicates a posterior probability exceeding 97.5%, accompanied by a 95% HPD that does not include 0, providing compelling evidence. **C**, AP negatively correlated with FTW under both low and high fatigue levels in Experiment 1. **D**, AP positively correlated with BTW under low fatigue in Experiment 1.

enhanced excitation-inhibition balance during low fatigue could heighten the brain's responsiveness to applied tACS and, at the same time, lead to more pronounced vulnerability in behavioral changes. Future studies could also benefit from incorporating objective measures of mental fatigue, such as physiological markers (Bafna and Hansen, 2021) or performance-based assessments. These objective measures can complement self-report scales, providing a more comprehensive understanding of fatigue dependence in tACS effects.

The implications of causality under low fatigue

In examining the causality identified under low fatigue, our study highlights that bottom-up processing, symbolized by FTW propagating from occipital to frontal areas (Alamia et al., 2023), exhibits a heightened sensitivity to visual contrast detection compared with top-down processing, denoted by BTW propagating from frontal to occipital areas, as FTW showed more robust correlations with VCT besides tACS effects on both FTW and BTW. This sensitivity could facilitate the accumulation of evidence (Pereira et al., 2021) during the prestimulus period, which subsequently impacts conscious detection (Baria et al., 2017; O'Connell and Kelly, 2021). Supporting this notion, primate studies, such as those by van Vugt et al. (2018), have shown that information from subtle stimuli might become attenuated

as it transits from the visual to the frontal cortex, particularly influenced by the prestimulus brain state. This hints at the crucial role of FTW in transmitting perceptual data throughout the cortex. Our findings, thereby, spotlight traveling waves as a compelling prospect for the neural correlates of consciousness and resonate with the global neuronal workspace theory in visual consciousness research (Mashour et al., 2020).

Furthermore, while prestimulus AP was not able to explain the modulation effects on VCT, it is closely associated with FTW and BTW, consistent with Alamia et al. (2023). This observation reinforced the notion that traveling waves, encompassing both temporal and spatial information, might offer a more comprehensive framework for understanding neural oscillations in visual perception (Zhang et al., 2018). Nevertheless, it is crucial to recognize the inherent limitations of our analysis of traveling waves, which was performed at the scalp level using EEG signals, subject to volume conduction effects. Key characteristics of traveling waves, such as their speed and the complex underlying neural networks, require more sophisticated recording methodologies with enhanced spatial resolution, such as intracranial EEG.

The potentials of 45/−45° phase-shifted tACS

To the best of our knowledge, this is one of the first studies to use 45/−45° phase-shifted tACS to concurrently modulate endogenous

traveling waves and their associated cognition in humans (Aleksichuk et al., 2019; Aksenov et al., 2023; Wischniewski et al., 2023b). Building on prior literature that emphasizes the significance of traveling waves in brain activity and cognition (Muller et al., 2018; Zhang et al., 2018; Halgren et al., 2019; Alamia et al., 2020), our findings accentuate the potential of 45/−45° phase-shifted tACS in both theoretical and clinical neuroscience realms. Future work can use 45/−45° phase-shifted tACS to modulate more cognitive functions which associate with cortical traveling waves. Furthermore, our results underscore two salient points: firstly, the criticality of considering brain states when appraising modulation effects, as highlighted previously, and secondly, the expansive scope that remains regarding the implications of 45/−45° phase-shifted tACS. Specifically, the observed modulation effects on FTW and BTW by 45° phase-shifted tACS imply that external propagating backward electric fields might offset endogenous traveling waves with the opposite direction (i.e., FTW was decreased), whereas they boost endogenous traveling waves with the same direction (i.e., BTW was increased). These effects, while promising, were modest, signaling the need for optimization. On the other hand, the external propagating forward electric fields induced by −45° phase-shifted tACS exhibited distinct impact as we expected on BTW and did not show influence in FTW or VCT. These findings highlight the intricate and potentially nonlinear interactions between external electrical fields, endogenous neural oscillations, and brain states, underscoring the need for further investigation into these complex dynamics.

Conclusion

In conclusion, our research reveals the potential causal role of prestimulus alpha traveling waves in visual contrast detection. The efficacy of 45° phase-shifted tACS in manipulating endogenous traveling waves and related cognitive functions highlights its promising applications in cognitive modulation and potential neurotherapeutic interventions.

Data Availability

The data and code to reproduce the results are available on the Open Science Framework at <https://osf.io/2ycdq>.

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